Arya Verma

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Education

University of Southern California Expected May 2027
B.S. Applied & Computational Mathematics *Los Angeles, CA*
- **Relevant Mathematics & Data Coursework**: Probability Theory; Mathematical Statistics; Numerical Methods; Fundamental Concepts of Analysis; Data Science with Python; Theory of Numbers.
- **Relevant Computer Science Coursework**: Data Structures and Object-Oriented Design; Discrete Methods in Computer Science; Analysis of Algorithms and Theory of Computing; Artificial Intelligence: Principles and Foundations.

Foothill College | Honors Institute Scholar 2023 – 2024
Computer Science, Mathematics, and Cognitive Science *Los Altos Hills, CA*
- **Relevant Coursework**: Linear Algebra; Differential Equations.

University of California, Santa Cruz 2022 – 2023
Computer Science *Santa Cruz, CA*
- **Relevant Coursework**: Programming Abstractions in Python; Computer Systems and Assembly Language.

Experience

Full Stack Engineer | Internship May 2026 – Aug. 2026
Tumeryk *Hybrid / Redwood Shores, CA*
- Engineered Tumeryk's Safari AI governance extension, rebuilding Chrome-specific policy controls; developed a test plan covering 50+ scenarios, conducted testing, and fixed redirect loops and attachment-blocking failures.
- Revamped Tumeryk's Shadow AI platform across backend and frontend, integrating Sophos DNS telemetry for AI discovery, usage monitoring, and governance; defined telemetry requirements and delivery formats and authored use-case briefs for executive-led sales and partner discussions.
- Designed and implemented bulk-import workflows across multiple platforms, enabling administrators to add AI tools and associated domains in batches.
- Developed CSV-driven Python regression tests for sensitive-data detection APIs, automating checks on entity types, detected text spans, confidence scores, and HTTP responses.
- Created a seven-part executive GenAI security lab with Immersive Labs, authoring content on AI architecture, threats, red teaming, and governance; audited internal product codebases to validate technical accuracy.

Full Stack Engineer | Internship May 2025 – Aug. 2025
Tumeryk *Hybrid / Redwood Shores, CA*
- Designed API contracts, protocol flows, and technical design documents for a dynamic AI guardrail policy service that adapts to user behavior and potential attack patterns.
- Investigated a critical RAG chatbot file-upload defect that blocked policy testing; documented root-cause findings and proposed DevOps-facing infrastructure fixes ahead of client deployment.
- Automated system-prompt generation and refined Topic Guideline guardrail UX flows, reducing false bug reports and improving the configuration experience for nontechnical users.
- Migrated product UI data visualizations from MUI X Charts to Recharts for improved maintainability and visual consistency; optimized staging procedures and reduced console error logs by approximately 80%.

Software Engineer | Internship Jun. 2023 – Sep. 2023
Kube8r Inc. *Hybrid / Dublin, CA*
- Built data-hosting workflows with AWS and MongoDB for application migration to a multi-cloud network; streamlined Kubernetes deployments and documented procedures for engineering handoff.

Software Engineer | Internship Jun. 2021 – Sep. 2021
Calance LLC *Remote / Anaheim, CA*
- Developed static web pages, researched UI/UX patterns for food delivery services, and prototyped AI voice-bot workflows with Amazon Lex and AWS Connect.

Projects

AI Search & Decision Systems | *Python, algorithms, dynamic programming* 2026
- Implemented BFS/DP rescue-robot planning, alpha-beta minimax game search, and value iteration for GridMDPs with deterministic tie-breaking and testable policy outputs.

Portfolio Optimization with Augmented Lagrangian Method | *Python, numerical methods* 2025
- Analyzed convergence behavior, computational cost, and perturbation sensitivity for constrained portfolio optimization experiments.

Technical Skills

Languages: Python, C++, Java, MATLAB, LaTeX
Tools & Platforms: Git, Docker, FastAPI, AWS, Kubernetes, MongoDB, Linux/Unix, Jupyter, NumPy, pandas, Matplotlib
Technical Areas: AI guardrails, sensitive data detection, regression testing, API testing, CSV/JSON parsing, log analysis, data cleaning, algorithms

Affiliations & Additional

USC AI Trust Hub: Student member, applied AI trust and evaluation.
Tutoring: Math Tutor, Foothill Pass The Torch.
Languages: Hindi (advanced), Spanish (intermediate).